

Jaeeun Kim

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EDUCATION

Purdue University

West Lafayette, IN

Ph.D. in Industrial Engineering

Expected Graduation: Dec. 2027

- **Research Focus:** Tactile Sensing, Loco-manipulation, Humanoid System, Reinforcement Learning
- **Relevant Coursework:** Robot Learning, Data Acquisition, Machine Design, Microcontrollers

Purdue University

West Lafayette, IN

Bachelor of Science in Robotics Engineering Technology

Dec. 2021

SKILLS

Hardware & PCB Design : KiCad, Eagle, Fusion, PCB Layout, Rapid Prototyping, 3D Printing, Soldering

Microcontrollers & Processing : ESP32, Jetson Nano Orin, Raspberry Pi, Arduino

Sensors & Protocols : Vision-based Tactile Sensor (GelSight), Visuo-tactile Sensor (3D-ViTac), SPI, I2C, CAN Bus

Software & Analysis : Python, MATLAB, C/C++, ROS2, MuJoCo, Data Visualization, Statistical Analysis

Engineering Experience & Projects

Humanoid Robot Tactile Integration & Control

Aug. 2023 – Present

Primary Developer

- Designed a small-scale humanoid robot system driven by ROS2, integrating 18 MG995 servos for complex loco-manipulation and 2 Betu 35kg digital servos for precise gripper actuation.
- Interfaced advanced tactile sensors into the robot's end-effectors and feet, establishing robust data pipelines using Raspberry Pi and Arduino.
- Processed and analyzed high-dimensional sensor data using Python to calibrate sensor outputs and optimize grasping algorithms.

Open-Source Mobile Robotics Architecture

Hardware & Software Developer

Oct. 2019 – Dec. 2021

- Designed and routed custom PCBs using KiCad and Eagle for an open-source mobile robot platform, overseeing schematic generation, component selection, and layout.
- Implemented SPI and I2C communication networks to synchronize motor controllers, power regulators, and peripheral sensor suites.
- Validated board designs through rigorous prototype testing, verifying voltage rails and debugging analog/digital circuits.

Academic Experience

Graduate Teaching Assistant - Vertically Integrated Projects

- **Mentored cross-functional research teams** in the development of a full-scale humanoid robot, IoT devices, and AI systems, successfully guiding projects from architecture definition through prototyping and validation.
- **Taught and troubleshot physics simulations (MuJoCo)** for robot learning applications, assisting teams in debugging software environments and resolving complex deployment issues.
- **Managed project timelines and milestones** for multidisciplinary teams, providing both technical direction and administrative support to ensure continuous progress and hardware de-risking.